

In class exercises: RI and assignment schemes

This exercise connects intuitions for randomization inference with the implications of different assignment schemes?

You have the following data:

ID	B	C	Z	Y
1	0	1	0	0
2	0	1	0	0
3	0	2	1	1
4	0	2	1	1
5	1	3	0	0
6	1	3	0	0
7	1	4	1	1
8	1	4	1	1

Calculate the p value for the sharp null hypothesis that effects of Z on Y are zero for all units under each of these conditions:

1. Z was assigned using **complete randomization**.
2. Z was assigned using **block randomization**, using B as a block.
3. Z was assigned using **cluster randomization**, where C denotes clusters.
4. Z was assigned using **block cluster randomization**, using B for blocks and C for clusters.

Implement one sided tests in all cases.

Discuss: we usually expect that blocking will improve power but clustering will reduce it. Is the logic in tension with what you just found. Explain why or why not.